From Bed Rest to Mars

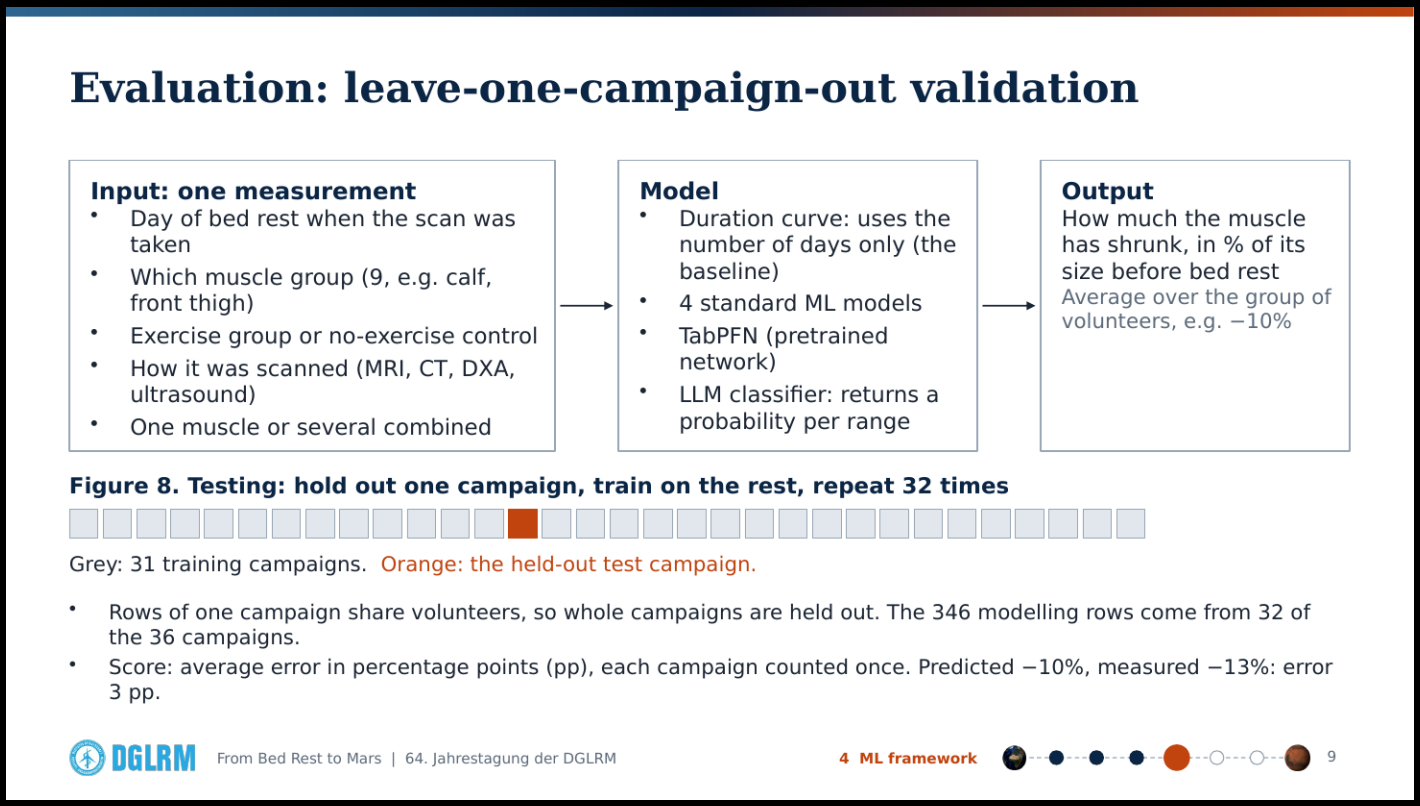
Practice script, Milad's part

Slides 9 to 14: about six minutes. Backup slides 15 to 21 are for questions only.

Amber italic lines are only for answering questions; do not read them out in the talk.

Slide 9 · Evaluation: leave-one-campaign-out validation

About one minute



Each row in our data is one measurement. It tells us how much one muscle shrank, on average, in one group of volunteers, on one day of bed rest. For every row, the models get five facts: the day of the scan, which muscle it is, whether the group did exercise or not, how the muscle was scanned, and whether it is one muscle or several together. The models must predict how many percent the muscle shrank compared with before bed rest.

We have 346 rows, but they come from only 32 separate studies, which we call campaigns. Rows from the same campaign come from the same people, so they are not truly separate. If a model trains on some rows of a campaign and is tested on others, it can cheat. So we hide one whole campaign, train on the other 31, and predict the hidden one. We repeat this 32 times, once for every campaign. The code checks that no campaign is ever on both sides.

We score each model by its average error in percentage points. Each campaign counts the same. As the simplest reference, we use a duration curve: it only knows how many days the bed rest lasted.

Slide 10 · LLM-based probabilistic classifier: input and output

About a minute and a half

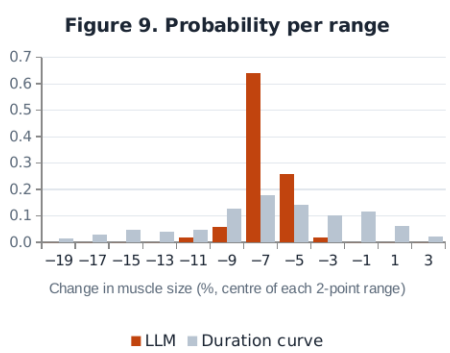
LLM-based probabilistic classifier: input and output

Request (one per measurement)

participants
men only, mean age 35, n = 11
protocol
head-down tilt bed rest, 21 days, control group
measurement
vastus lateralis, MRI cross-sectional area
target
day 21 of bed rest
typical_curve_other_campaigns
duration curve fitted on the other 31 campaigns
observations_other_campaigns
rows from the other 31 campaigns, most similar first, up to about 26,000 tokens
question
Which range will the change fall in? 19 options: below −30%, 2-point ranges up to +4%, +4% and above

LLM classifier
fixed version

A classifier: it returns one probability per listed option and never writes text.



- Prediction −6.6%: range midpoints weighted by probability (open ends −31% and +5%)
- Measured −9.4%: error 2.8 pp, the median error

Never sent, checked on every request: the held-out campaign's measurements, and any paper, author or campaign name. All 430 answers are saved, and every result rebuilds from them.

The normal models only see numbers in a table. For the LLM, we wrote each measurement out in words. We used a special kind of LLM. It does not chat or write text. It gets a description and a fixed list of answers, and it gives back a probability for each answer.

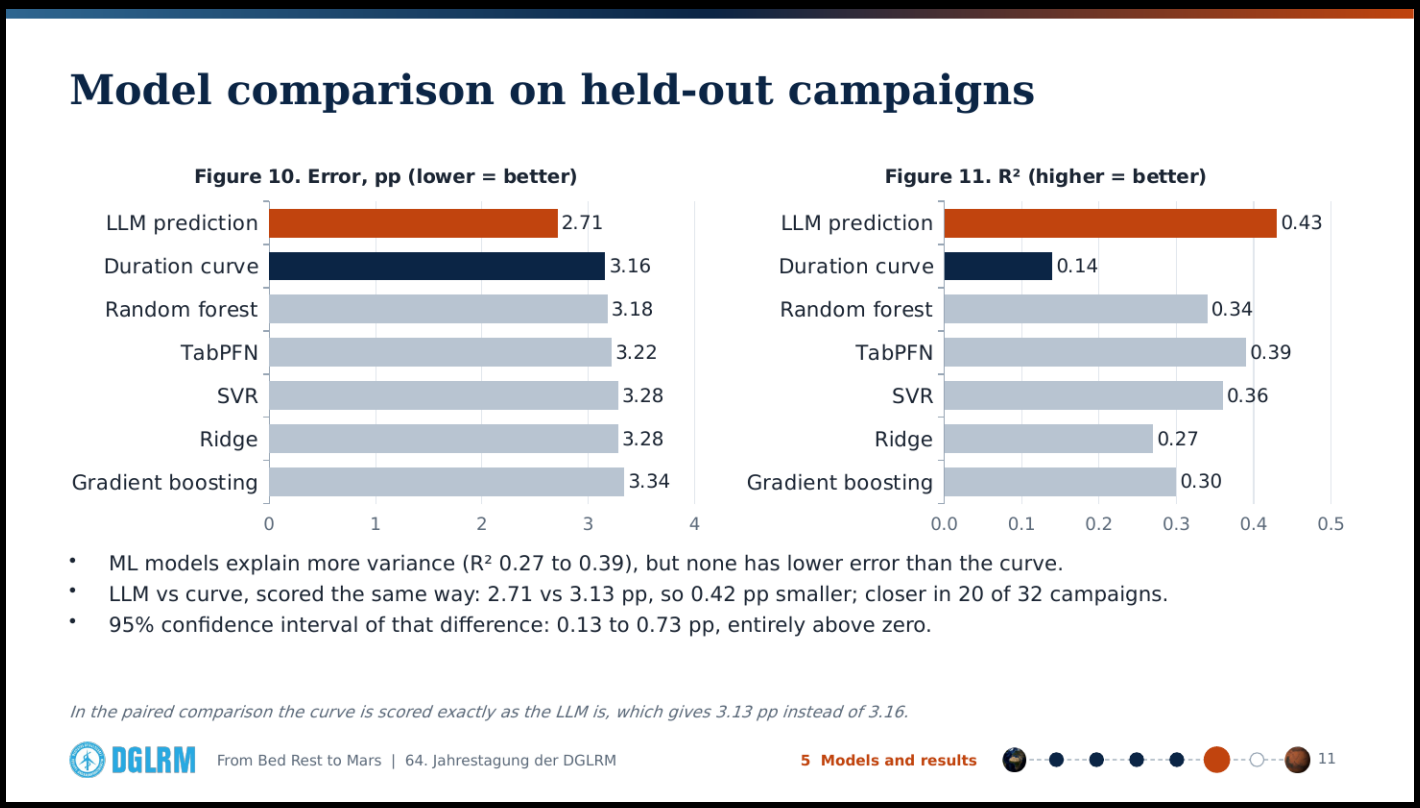
On the left is a real request. It describes the volunteers, the bed-rest study, the muscle, how it was scanned, and the day. It also includes the duration curve and the most similar rows from the other 31 campaigns. Then it asks: in which of these 19 ranges will the change fall?

On the right is the answer: a probability for each range. We never ask the model for a number. We turn the probabilities into one number by taking a weighted average of the middle of each range. Here that gives −6.6 percent. The real value was −9.4 percent. This example has a typical error for this model.

The request never includes any data from the hidden campaign, and never the name of a paper, author or campaign. The code checks this for every request. We saved all 430 answers, so anyone can check our results without running the model again.

Slide 11 · Model comparison on held-out campaigns

About a minute and a half



We tested four standard machine learning models: ridge regression, random forest, support vector regression and gradient boosting. We also tested TabPFN, a network that was trained in advance on many small tables.

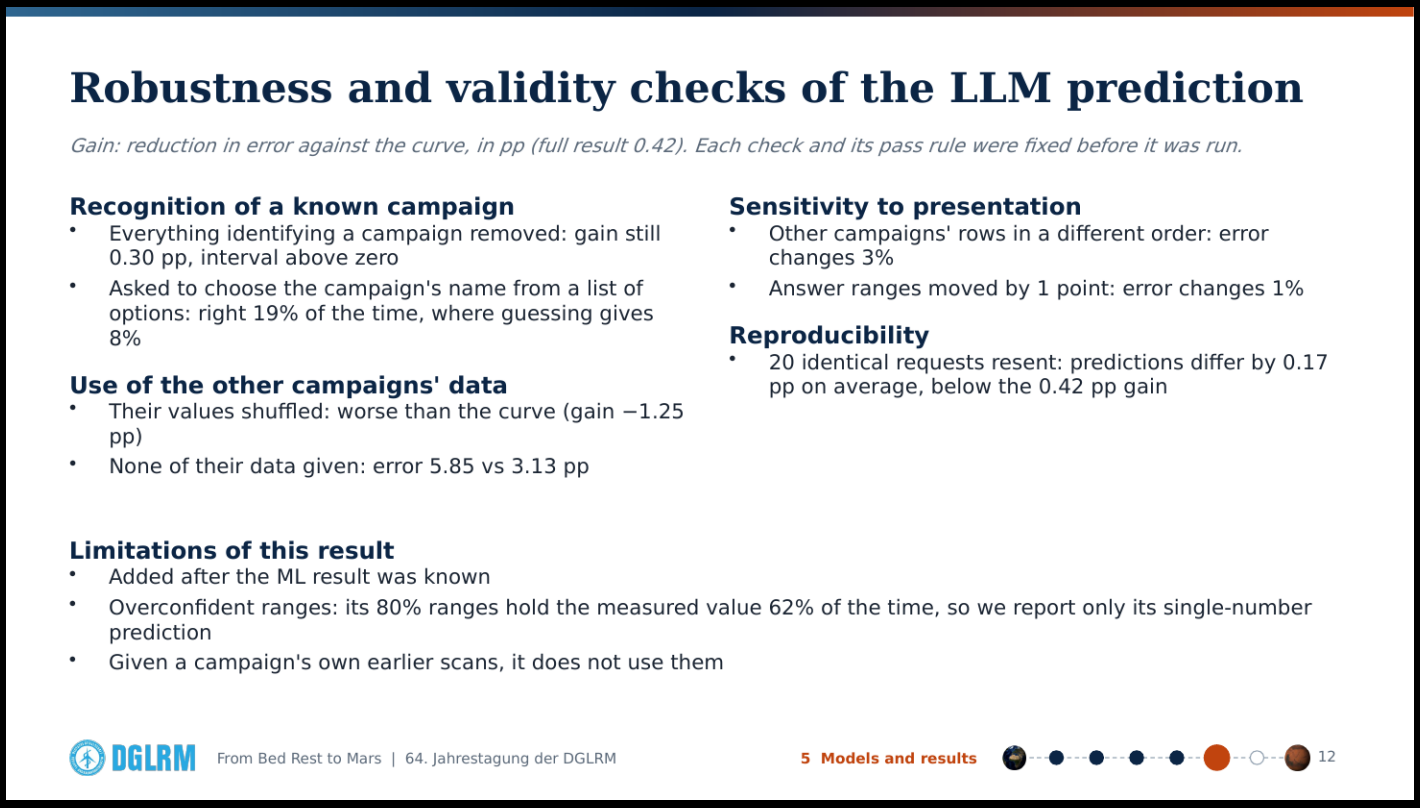
The left chart shows the error on the hidden campaigns. Lower is better. The simple duration curve is off by 3.16 percentage points. The best machine learning model, the random forest, is off by 3.18. So none of them beats the simple curve. The right chart shows R^2, which is how much of the differences between rows a model explains. Here the machine learning models do better, because they learn that some muscles shrink more than others. But most of the error comes from differences between whole campaigns, such as the scanner, the people or the study design. None of our inputs captures that. We had decided in advance to report this result either way.

The LLM is at the top. Its error is 2.71 percentage points, the lowest of all, and it also explains the most, with an R^2 of 0.43. When we compare it with the curve on exactly the same rows, the error drops from 3.13 to 2.71. That is 0.42 points better. The 95 percent confidence interval goes from 0.13 to 0.73. Even the low end is above zero, so the improvement is real. The LLM beats the curve in 20 of the 32 campaigns. No other model has a clear improvement like this.

If asked about the target: we had set a goal of 15 percent improvement in advance. The error improved by 13.3 percent and the probability score by 17.7 percent.

Slide 12 · Robustness and validity checks of the LLM prediction

About a minute and a quarter



We wrote down every check and its pass rule before running it. First, does the model just remember the studies? It may have read these papers during training. When we remove every detail that could identify a campaign, the LLM is still clearly better, by 0.30 points. We also asked it to pick the campaign name from a list. It was right 19 percent of the time, while random guessing gives 8 percent.

If asked about recognition: of the three campaigns behind most of the improvement, it recognised only one.

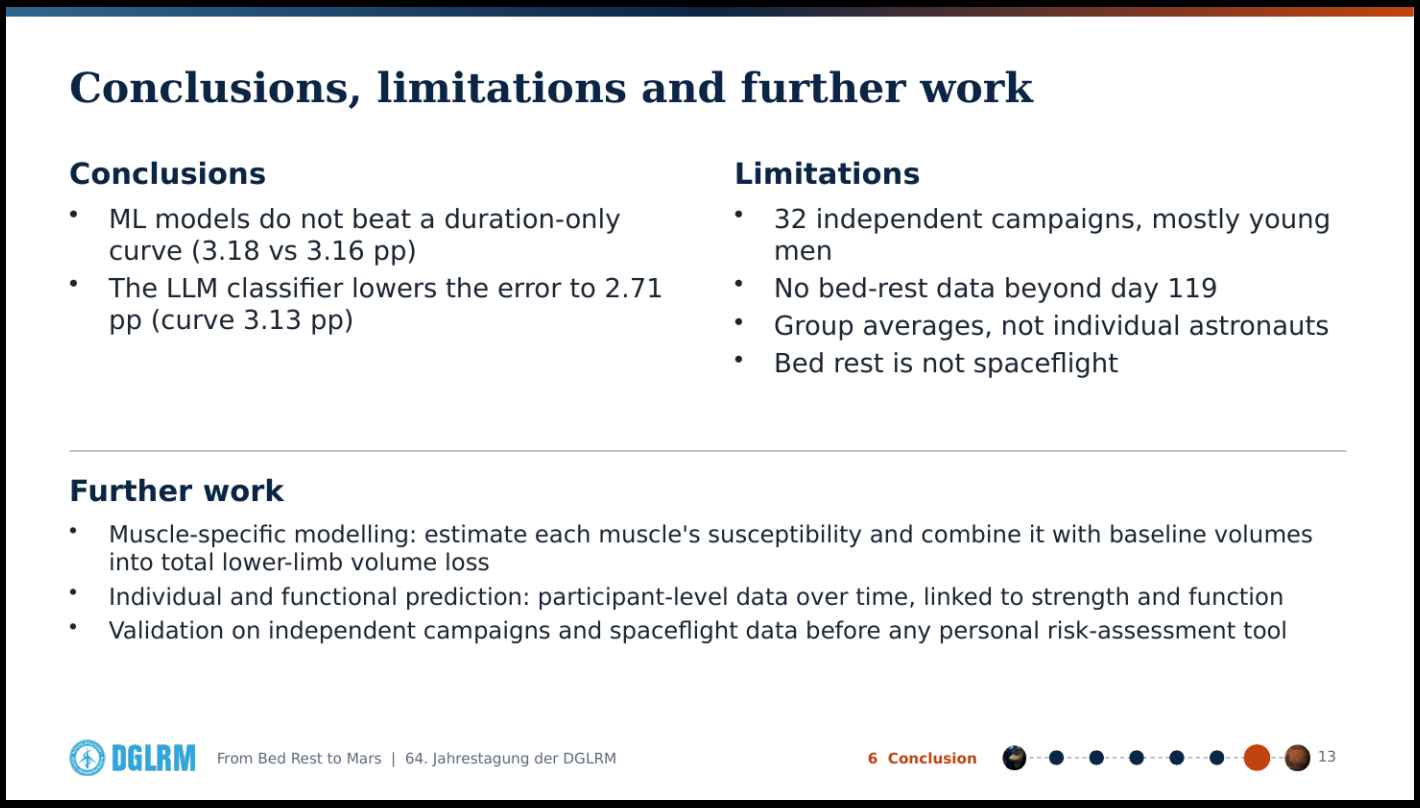
Second, does it really use the data from the other campaigns? When we mix up those values, it becomes 1.25 points worse than the curve. With no data at all, the error jumps to 5.85. So the improvement comes from reading that data.

Third, does the format matter? Changing the order of the rows or moving the ranges changes the error by only 1 to 3 percent. Fourth, is it stable? Sending the same 20 requests again changes the predictions by 0.17 points on average, smaller than the 0.42 improvement.

There are three limits. We added the LLM after we had seen the machine learning results. Its 80 percent ranges contain the real value only 62 percent of the time, so it is too confident and we only use its single number. And when we give it earlier scans from the same campaign, it ignores them.

Slide 13 · Conclusions, limitations and further work

About forty-five seconds



To sum up: with only 32 campaigns, standard machine learning models do not beat a simple curve that only knows the number of days. The LLM classifier, which reads the description of each campaign and the data of the others, lowers the error from 3.13 to 2.71 percentage points.

The main limits: we have only 32 campaigns, mostly young men, and no bed-rest data after day 119. Every row is a group average. So we predict how much a group loses on average, not the risk for one astronaut.

Next steps. First, find out how strongly each muscle reacts, and use that with normal muscle sizes to estimate the total loss in the legs. Second, use data from individual people over time, and link muscle loss to strength and function. Third, test everything on new campaigns and on real spaceflight data before building any personal risk tool.

If asked about the first step: the dataset has 30 group time points where three or more muscles were measured together, across 14 campaigns.

Slide 14 · Thank you for listening

Closing



Thank you for listening. We are happy to answer your questions.

Backup slides: 15 and 16 on the data and the search, 17 and 18 on part 1, 19 to 21 on part 2.

Slide 15 · Backup: dataset coverage and limitations

Backup: only if asked

Backup: dataset coverage and limitations

Did you miss studies?

- Four sources searched from 2013 on, plus 10 older or open-data studies (sources and queries on the next slide)
- Not covered: Embase (no access)

One campaign dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each campaign one vote; the re-run without it is still to do

Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

If asked about extra checks: three planned checks have not been run yet. These are the composite-first analysis, the analysis without the MEDES campaign, and the analysis with MRI only. Only the weighting check has been run, and it changed the results very little.

Slide 16 · Backup: literature search sources and query

Backup: only if asked

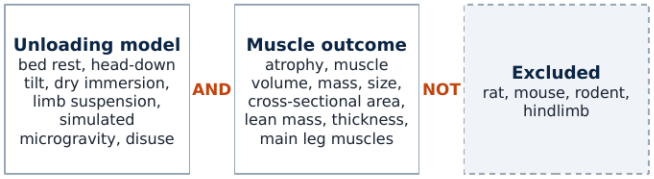
Backup: literature search sources and query

Table 3. Sources (searched 4 September 2026)

Source	Records
PubMed	1,412
Scopus	1,757
Web of Science Core Collection	1,876
NASA Technical Reports Server	686
Earlier studies and NASA open data	10
Total	**5,741**

- Journal databases: publications from 2013 onwards
- Older work enters through studies we held before the search (9) and one NASA open-data campaign
- NASA reports: five short keyword queries, e.g. "bed rest" "muscle volume"
- Not searched: Embase (no institutional access), Cochrane CENTRAL, trial registries, citation chasing

Figure 12. Query: two required blocks



Web of Science string, verbatim

```
TS=(("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR
  OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"
  OR "limb suspension" OR ULLS OR "simulated microgravity"
  OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"
  OR "muscle unloading" OR disuse)
 AND
 (atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"
  OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"
  OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius
  OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```

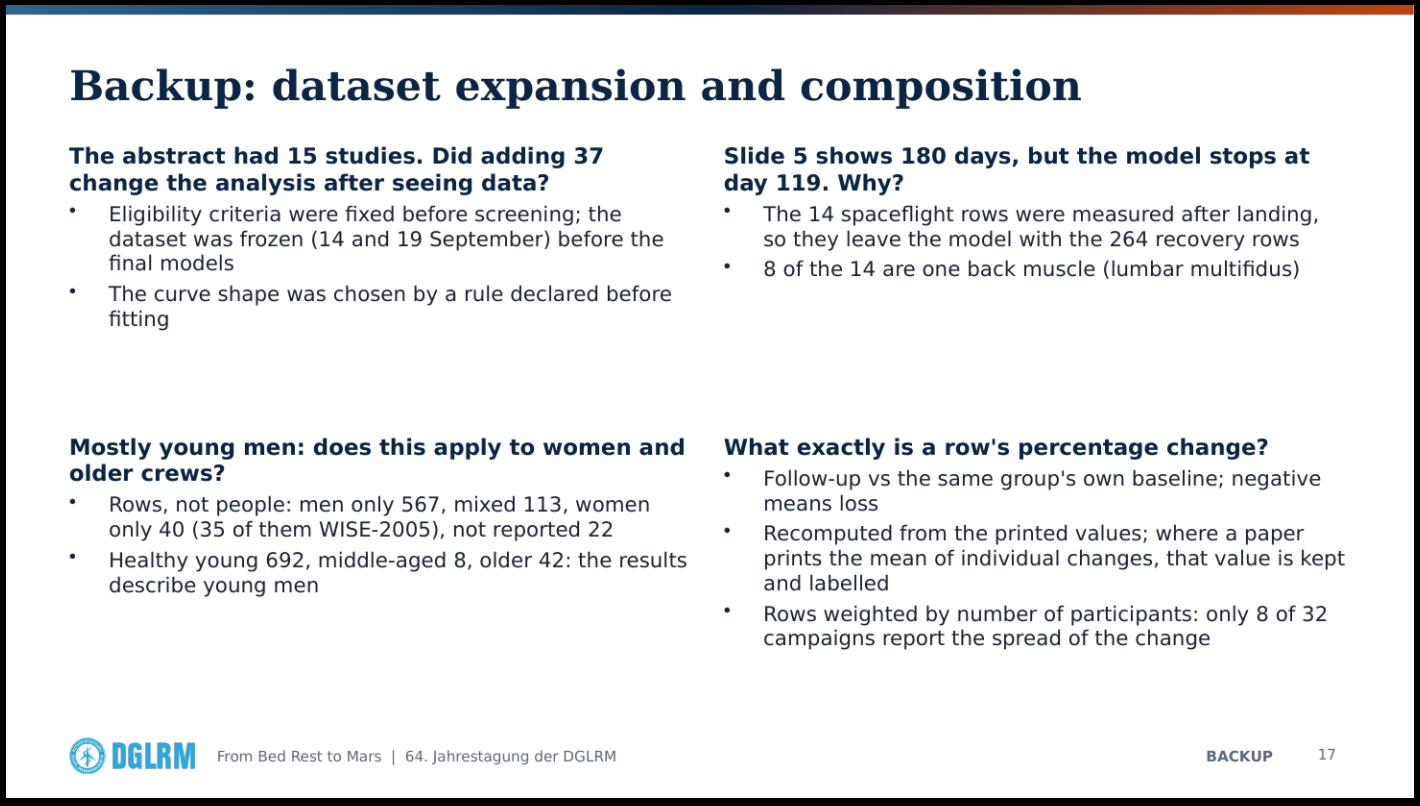
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All four search strings are in the appendix of the report and in docs/literature-review/search_log.md. PubMed uses the same two blocks, with MeSH terms and a filter for human studies. Scopus and Web of Science remove animal studies in the search itself.

The main weakness is that we only searched papers from 2013 onwards. Only two of ten known modelling papers fall in that period, and we did not test the search without this limit. We could not search Embase because we had no access. Embase includes conference abstracts that the other databases miss.

Slide 17 · Backup: dataset expansion and composition

Backup: only if asked · Part 1 (Niloufar)



We set the rules for which studies to include, in Table 2 of the report, before we started screening. Version 1.0 of the dataset was frozen on 14 September and version 1.1 on 19 September. All results use version 1.1.

We use the saturating curve unless another shape fits better by at least 4 AIC points.

The spaceflight rows are still in the dataset, marked as spaceflight. They were measured after landing, so the filter for measurements during unloading leaves them out.

Weighting rows by their uncertainty would need the spread of each change, and most papers do not report it. When we compared both ways of weighting, the effect of duration changed by less than 1 percentage point.

Slide 18 · Backup: certainty of the muscle-group curves

Backup: only if asked · Part 1 (Niloufar)

Backup: certainty of the muscle-group curves

Is each line fitted to its own muscle?

- No. The day-60 values are fitted per muscle group (named muscles, 25 campaigns); the lines share one time course (τ = 60 days)
- Separate shapes cannot be estimated: six of nine groups rest on four campaigns or fewer

Why a saturating curve?

- Logarithmic, saturating and spline curves fit equally well (within 1.3 AIC points): the loss slows, but the data cannot say along which curve
- By day 119 the curve reaches 86% of its plateau, −17.3% (95% CI −20.9 to −13.7)

How certain are the day-60 values?

- Calf −15.8% (95% CI −18.9 to −12.7); front thigh −10.1% (−13.6 to −6.6); hip rotators −1.9% (−3.9 to 0.2)
- Calf vs shin, the ankle's opposing pair: 5.6 pp more loss (2.1 to 9.1), p = 0.003
- Hip rotators and outer hip: one campaign each

Is the calf result an artefact of how often it is scanned?

- The order holds after adjusting for duration, countermeasure group and imaging method
- Calf: 78 rows from 13 campaigns, the second best-covered group after the front thigh (86 rows, 22 campaigns)

The dots at day 60 on slide 8 are the estimates for each muscle group, with the ranges shown here. All lines use the same curve shape, scaled to these dots. So they show the shape we assume, not a separate fit for each muscle.

The ranges are 95 percent intervals that take into account that several rows come from the same campaign. We show the calf versus shin comparison on its own because it does not depend on which muscle we use as the reference.

Slide 19 · Backup: significance and relevance of the LLM gain

Backup: only if asked

Backup: significance and relevance of the LLM gain

Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small; we present it as a secondary finding

20 of 32 campaigns is close to a coin flip

- Counting wins only: p = 0.22 (sign test)*
- Using the size of each win: mean gain 0.42 pp, 95% CI 0.13 to 0.73

Does it rest on a few campaigns?

- 3 long MRI campaigns (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI −0.01 to 0.43*
- Only SPRINT is recognised, and with campaign details removed the LLM still wins there (3.79 vs 5.92 pp)

Did it read these papers in training? Was it added after the fact?

- No test can rule out training exposure; hiding campaign details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

** Our calculation from the saved LLM predictions; not in the report.*

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The two numbers with an asterisk are our own calculation from results/forecast_predictions.csv. They are not in the report.

If we remove the three campaigns that give most of the improvement, the interval touches zero. Say this openly if asked.

Slide 20 · Backup: ML results and fairness of the comparison

Backup: only if asked



The open question is the nearest-neighbour test. We do not yet know whether a simple method that just picks similar rows would do as well as the LLM.

The 271 men-only rows and 100 countermeasure rows are counted on the 346 modelling rows. The report makes the same point with the 425 leg rows before duplicates were removed: 347 men-only and 126 countermeasure rows.

Slide 21 · Backup: applicability to Mars mission planning

Backup: only if asked

Backup: applicability to Mars mission planning

A Mars transit is about 180 days; the data stop at 119

- Only one campaign reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average

Within 5 to 119 days, and for groups, the data support three points for mission planning. Most of the loss happens in the first weeks. The calf muscles need protection first. Groups with a countermeasure, such as exercise, lose less muscle.